

The Effect of Commute Time on Wage Income

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Abstract

Recent studies show a positive relationship between commute time and wage income, increasingly relevant with the shift in the labor market towards remote work. This paper examines that relationship using 2024 IPUMS data on over 1.3 million U.S. workers and an OLS regression model that controls for demographics, education, race, and hours worked. The results show that an increase of 10 minutes of commute time is associated with a 1.2% increase in wage income. The effect does shrink with controls but remains statistically significant, suggesting workers are compensated for commuting costs, relevant to remote work and transit policy.

Section I: Introduction

This paper analyzes the effect of commute time on average wage income among United States workers in 2024. Commute time is a real cost that is incurred by those in the workforce, whether that be time, transportation cost, or even stress. This question is important because it is crucial to understanding how the labor market compensates workers now more than ever as remote work has skyrocketed since the 2020 pandemic. This paper expects a positive relationship between commute time and wage income, because those who are willing to commute longer have wider access to higher paying jobs, and the labor market should compensate them for the extra costs.

Research has been conducted on how commute time can influence workers' income. French, Popovici, and Timming (2020) found that longer commute times are associated with higher annual earnings, suggesting that workers who are willing to commute further may have access to higher paying jobs. Troncoso, de Grange, and Rodríguez (2021) found that commute time could explain a meaningful portion of the gender wage gap in Chile, showing that commute is an important factor in determining income. Looking at both studies, there is significant evidence that commute time plays a factor in the labor market, providing substantial evidence that recent U.S. data will show a similar relationship.

Using data on 1,392,789 employed U.S. adults ages 18-65 in 2024, this analysis finds that longer commute times are associated with higher wage income. For every 10 minutes added to commute, there is a 3.1% increase in annual wage income in the baseline model. However, the coefficient falls to 1.2% once demographics, education, race, and hours worked are controlled for. Even with these other factors, the relationship between commute time and wage income

remains positive and statistically significant, concluding that commute time does influence wage income.

Section II: Data Summary

The data source used for this research project is IPUMS USA (2024), which uses census and survey data, with individuals as the unit of observation. Since the research question only focuses on those in the workforce, only individuals between the ages of 18–65 were kept. Individuals with an annual wage income of zero, or with missing values were dropped from the sample. Table 1 defines the key variables used, and Table 2 provides the summary statistics for the full sample. Figure 1 shows the relationship between commute time and wage income, showing a positive correlation: as commute time rises, wage and salary income tends to rise with it. This graph is consistent with the regression model.

Table 1. Variable Descriptions

Variables	Description
ln_wage	Natural log of annual wage income
commute10	One way commute time divided by 10
age	Individual's age
female	Whether individual is female
educ	Highest level of education attainment categorical variable
race	Individual's Race
uhrswork	Usual hours worked per week

Figure 1. Scatter Plot

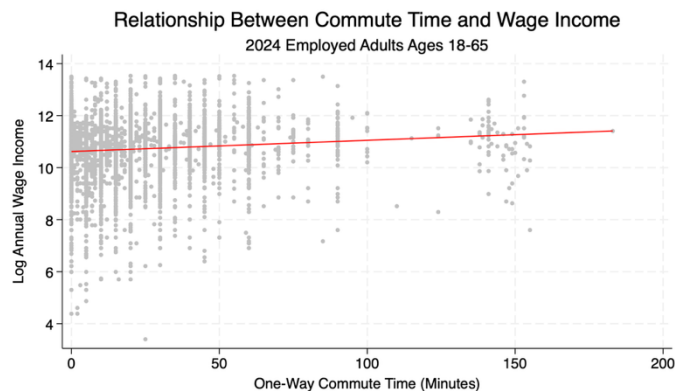


Table 2. Summary Statistics

	N	Mean	Std. Dev.	Min	Max
Log Annual Wage Income	1392789	10.728	1.088	1.386	13.718
Commute Time, 10-Minute Units	1392789	2.349	2.366	0.000	19.500
Age	1392789	41.580	13.272	18.000	65.000
Female	1392789	0.486	0.500	0.000	1.000
Usual Weekly Work Hours	1392789	39.324	11.518	1.000	99.000

Section III: Econometric Model

$$\ln(Wage_i) = \beta_0 + \beta_1 Commute10_i + \beta_2 Female_i + \beta_3 (Commute10_i \times Female_i) + \beta_4 Age_i + \beta_5 High\ School_i + \beta_6 Some\ College_i + \beta_7 Bachelor's\ or\ Higher_i + \beta_8 BlackRace_i + \beta_9 Asian_i + \beta_{10} Other\ or\ Multiracial_i + \beta_{11} WeeklyHoursWorked_i + \varepsilon_i$$

In this model, the dependent variable is the natural log of annual wage. This variable is logged because income is highly skewed and logging also helps address heteroskedasticity. This also allows the coefficient to be interpreted as a percent change in annual wage income. The main independent variable is commuting time, measured in 10-minute units, which makes the coefficient easier to interpret than a one-minute change would.

There are five control variables included: age, gender, education, race, and usual weekly work hours. Age is included since those who are older and have more experience are more likely to make more money than someone who is just starting their career. Gender is included to control for wage differences between men and women. Education is included because workers with higher education are generally expected to make more money. Race is included to account for demographic differences in earnings. Both education and race are dummy variables because they are categorical, with the lowest category in each, no high school diploma for education and White for race, acting as the reference group. Lastly, usual weekly work hours are included because individuals who work more hours are likely to earn a higher income.

There is also an interaction variable between commute time and the female variable. This interaction tests whether the relationship between commute time and wage income is different for women than for men. Without the interaction term, the model would assume that the effect of commute time on wages is the same for both genders, even if it isn't.

Section 4: Empirical Results & Discussion

VARIABLES	(1) OLS	(2) + Demographics	(3) + Education	(4) + Race	(5) + Hours & Interaction
Commute Time	0.031*** (0.000)	0.022*** (0.000)	0.027*** (0.000)	0.028*** (0.000)	0.012*** (0.000)
Female		-0.330*** (0.002)	-0.398*** (0.002)	-0.394*** (0.002)	-0.220*** (0.002)
Commute Time x Female					0.004*** (0.001)
Age		0.024*** (0.000)	0.022*** (0.000)	0.022*** (0.000)	0.016*** (0.000)
Education = 2, High School			0.272*** (0.004)	0.258*** (0.004)	0.202*** (0.004)
Education = 3, Some College			0.450*** (0.004)	0.432*** (0.004)	0.374*** (0.004)
Education = 4, Bachelor's or Higher			1.095*** (0.004)	1.066*** (0.004)	0.896*** (0.004)
Race = 2, Black				-0.194*** (0.003)	-0.171*** (0.003)
Race = 3, Asian				0.016*** (0.003)	0.078*** (0.003)
Race = 4, Other or Multiracial				-0.077*** (0.002)	-0.070*** (0.002)
Weekly Work Hours					0.042*** (0.000)
Constant	10.655*** (0.001)	9.854*** (0.003)	9.297*** (0.005)	9.349*** (0.005)	7.958*** (0.006)
Observations	1,392,789	1,392,789	1,392,789	1,392,789	1,392,789
R-squared	0.005	0.110	0.240	0.243	0.426
Demographic Controls	No	Yes	Yes	Yes	Yes
Education Controls	No	No	Yes	Yes	Yes
Race Controls	No	No	No	Yes	Yes
Hours Worked	No	No	No	No	Yes
Interaction	No	No	No	No	Yes

Robust standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1

Table 3 presents the simple OLS results with the progression of adding controls and one interaction variable. In Column 1, the coefficient on commute time is 0.031 and is statistically significant at the 1% level. This means that holding everything else constant, for every 10 minutes of commute time, wage income increases by 3.1%. This significance indicates that commute time is associated with different levels of wage income prior to any other factors being added.

Once demographic, education, race, and hours worked controls are added, the coefficient on commute falls to 0.012 in Column 5. This means that in the final model, holding everything else constant, every 10 minutes of commute time is associated with a 1.2% increase in wage income. Although the effect is smaller than the baseline model, it remains statistically significant at the 1% level. This shows that commute time still has a meaningful and independent effect on wages even after adding education, race, age, gender, and hours worked.

This drop suggests that the baseline model suffered from omitted variable bias. When the controls were added, the correlation that Column 1 said was solely commute time decreased. However, even though the variables controlled for reduced the correlation, the effect does not disappear and remains statistically significant.

The table also highlights a consistent pattern across the Female category. The Female coefficient is negative throughout, starting at -0.330 in Column 2 and slightly increasing to 0.220. This shows the presence of the gender wage gap even after controlling for the other variables. Additionally, the interaction term between Females and commute time is positive and significant, demonstrating a higher return on commute time for women than for men. This suggests that a longer commute for women may slightly reduce the gender pay gap.

The R-squared also increases substantially across each column, from 0.005 in the OLS model to 0.426 in the full model, showing that the added controls explain a meaningful share of the variation in wages. This also supports the earlier point that the baseline model was missing important variables. One unexpected result is the coefficient on Asian, which started at 0.016 and increased dramatically to 0.078 once hours worked is included in Column 5. This is because Asian workers in the sample work slightly fewer hours than White workers, which was hiding part of their wage advantage until hours worked was controlled for, meaning the earlier models weren't telling the full story on the wage difference for Asian workers. In conclusion, as more control variables and an interaction term are added, the coefficient on commute time drops but stays statistically significant, showing that commute time has a real effect on wages.

Section 5: Conclusion, Limitations & Future Work

Longer commute times are associated with higher wage income, even after controls. This reflects that when a worker is willing to put in more time and effort, they are being compensated for that. It could also represent that better paying jobs are located further away from home locations. This could also mean that people who make more money are more willing to commute in order to live in a bigger house or a nicer location. Additionally, women show a slightly higher return than men when it comes to commuting, which does relate to the gender wage gap.

Although control variables were added to account for omitted variable bias, other relevant variables were not available in this dataset, including industry type, cost of living, and skill set. These additional factors likely correlate with both commute time and wage income, which could leave some endogeneity in the model. As a result, the coefficient on commute time may still partly reflect these unobserved factors rather than the true effect of commuting itself. Industry

type might also play a larger role than the others, since certain high paying industries may also require longer commutes.

For future research, taking data from the census and looking at it state by state and city by city might be able to tell a better picture. The United States is incredibly diverse, and results at the national level might not reflect what's happening depending on where you live. Being able to understand commute times and income on a state level would greatly improve the research.

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